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Transferring deep learning models for hydrographic feature extraction from IfSAR data in Alaska

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Abstract:

The National Hydrography Dataset (NHD) managed by the U.S. Geological Survey (USGS) is a database of vector geospatial features representing the surface water features across the United States. It provides a common baseline of semantic hydrography content that can be shared among organizations for managing and evaluating water and associated natural resources, as well as for applied hydrologic uses. The NHD was originally compiled from blue-line hydrographic content on USGS topographic maps (1947 to 1992, with associated edits and refinement through 2022). The USGS 3D Elevation Program (3DEP) provides high-accuracy, high-resolution (HR) elevation data for the United States (5-m resolution in Alaska and 1 m resolution elsewhere). The NHD is being updated with higher-quality feature representations through efforts that derive hydrography from 3DEP HR elevation datasets. Deriving hydrography from elevation through traditional flow routing and interactive methods is a complex, time-consuming process that must be tailored for different hydrogeomorphic conditions. The large volume of surface water features and HR remote sensing data make manual annotation of the water features over the entire Nation infeasible. Furthermore, annual and seasonal variations of surface waters warrant some level of periodic updates to hydrography. Advances in deep learning technologies provide an opportunity to automate hydrography extraction and scale up the process to a nationwide level. One major challenge, however, is the effect of spatial heterogeneity due to the wide variety of hydrogeomorphic conditions in the United States. In other words, it is unclear how a deep learning model pre-trained in one set of hydrogeomorphic conditions can be effectively applied to other conditions for hydrographic feature extraction.

This paper aims to provide some clarity in this regard by testing automated deep learning and its transferability to the extraction of hydrography from digital elevation model (DEM) data spanning a range of hydrogeomorphic conditions in Alaska. In transfer learning, the knowledge (e.g., neural network weights) from one domain is transferred to other domains. It helps decrease training requirements in the target domain, such as fine-tuning a model with fewer training samples and less processing time than training from scratch (Kimura et al., 2020).

This work examines whether applying transfer learning methods to extract hydrography from 5-m resolution DEM data derived from Interferometric synthetic aperture radar (IfSAR) can improve machine learning predictions. Reference data consist of the best available hydrography, which is derived through flow-routing techniques applied to the available IfSAR data, along with manual editing and interpretation of high-resolution satellite image data.

It has been demonstrated that the deep convolutional neural network (CNN), U-Net, can be successfully trained to extract hydrography from IfSAR (Stanislawski et al., 2021), and Stanislawski et al. (2022) began testing transfer learning methods using the U-net model for watersheds adjacent to the Alaska study area (to help scale up hydrography predictions for a larger Alaska region). Research presented here extends transfer learning tests to a broader range of hydrogeomorphic conditions in Alaska, where the quality of the IfSAR data and the available reference data may vary substantially from earlier test data.

Numerous U-net models were trained to extract hydrography from several IfSAR-derived layers for a 50-watershed study area (~4,600 square km) in north-central Alaska (Figure 1a). The best of the models tested from a range of U-net

model configurations and training scenarios uses 3,600 training windows of 128x128 pixels distributed over each of the 12 training watersheds in the study area (Figure 1b), along with 10 input layers. The average F1-score resulting from the best model for the 38 test watersheds is 72 percent for unfiltered data and 80 percent for filtered reference data, which removes prospective errors based on modeled surface water depth estimates. The best model is planned to be applied to predict hydrography for over 1,000 watersheds in various geographic conditions within Alaska, where similar reference data are available for comparison. Average F1 scores for watersheds within 8-digit hydrologic units (HU8) (U.S. Geological Survey, 2023) can be compared for model predictions determined with and without transfer learning (Figure 1a).

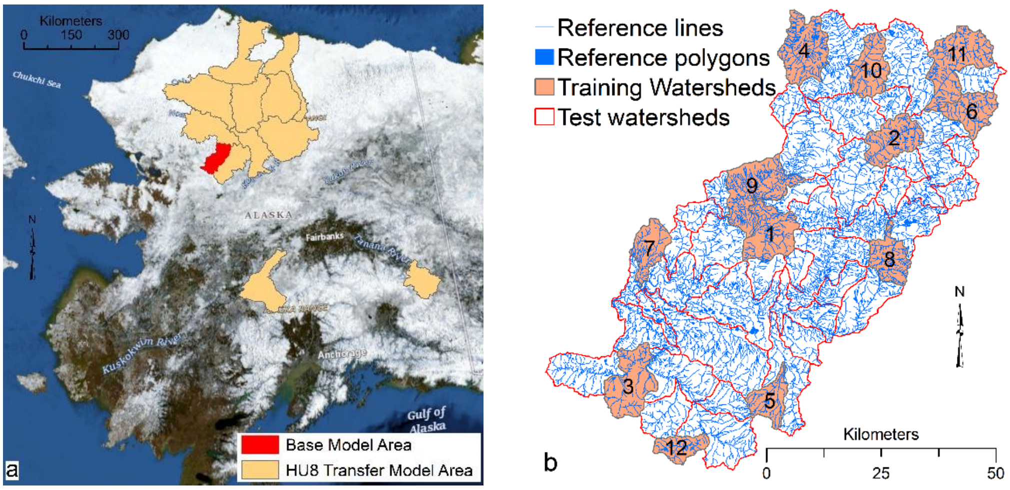

Figure 1. (a) Study area in Alaska used for base model and transfer learning with 8-digit hydrologic unit (HU8) areas. (b) 12 training watersheds, 38 test watersheds, and reference hydrography data used for base U-net model.

When transfer learning is applied between different hydrogeomorphic conditions, additional training is performed on the base (best) model to refine it for conditions in each of the target HU8 areas. The premise of transfer learning is that the initial layers of a deep neural network model encapsulate a substantial amount of general knowledge, whereas the later dense layers are for specific local details and for the classification of extracted features (Yosinski et al., 2014). Research by Shirokikh et al. (2020) indicates that fine-tuning the first layers of U-net models outperforms fine-tuning the last layers in transfer learning tests when used to segment Magnetic Resonance Imaging (MRI) data. Building on those results, in the current paper, the base model can be transferred to target areas using this general strategy: (1) load the base model weights, (2) freeze some set of layers in the model, (3) use one or more watersheds of reference data in a target HU8 to train the unfrozen layers to refine the model for the target area, and (4) unfreeze all layers in the model and retrain the model with a much-reduced learning rate to further refine the model. Several variations of this strategy are planned to be tested to refine the base model for the target areas. Terrain conditions within each HU8 watershed can be summarized and compared with the conditions in the base model area, and the accuracy and suitability of methods can be assessed.

Disclaimer

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